ELEC4631 Solutions Tutorial 4

1. (a) Recall from Tutorial 2 Q5 that the control canonical form is:

$$S_{\text{control}} = \left(\begin{bmatrix} 0 & 1 \\ 3 & -2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 & 1 \end{bmatrix}, 0 \right).$$

Since $y = Cx(t)$ ($D = 0$) it follows that

$$y^{(1)} = C\dot{x} = C(Ax + Bu) = CAx + CBu.$$

Therefore, $y^{(1)}(0) = CAx_0 + CBu(0)$.
Let $x_0 = \begin{bmatrix} x_{10} & x_{20} \end{bmatrix}^T$. $y(0) = 0 = \begin{bmatrix} -1 & 1 \end{bmatrix} x_0$ and $y^{(1)}(0) = 1 = \begin{bmatrix} 3 & -3 \end{bmatrix} x_0 + u(0)$, giving two equations $-x_{10} + x_{20} = 0$ and $3x_{10} - 3x_{20} = 1 - u(0)$. The second equation is equivalent to $-x_{10} + x_{20} = (u(0) - 1)/3$. Obviously, whenever $u(0) \neq 1$, there is no solution (x_{10}, x_{20}) that will satisfy this last equality together with the equality $-x_{10} + x_{20} = 0$.

(b) In the observer canonical form,

$$S_{\text{observer}} = \left(\begin{bmatrix} 0 & 3 \\ 1 & -2 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \end{bmatrix}, 0 \right).$$

Since $y^{(1)}(0) = CAx_0 + CBu(0)$, x_0 needs to satisfy the equation (using $y(0) = 0$ and $y^{(1)}(0) = 1$)

$$\begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} x_{10} \\ x_{20} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 - u(0) \end{bmatrix},$$

with solution $x_{10} = 1 - u(0)$ and $x_{20} = 0$.

2. The matrix

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -2 \end{bmatrix}$$

has two complex eigenvalues $-1 \pm \jmath$ and can be decomposed as

$$A = V \text{diag}(-1 + \jmath, -1 - \jmath) V^{-1}$$

with

$$V = \begin{bmatrix} 1 & 1 \\ -1+\jmath & -1-\jmath \end{bmatrix}$$

Then

$$\begin{aligned}
e^{At} &= V\operatorname{diag}(e^{(-1+\jmath)t}, e^{(-1-\jmath)t})V^{-1}, \\
&= -\frac{1}{2\jmath}e^{-t}\begin{bmatrix} -2\jmath\sin t - 2\jmath\cos t & -2\jmath\sin t \\ 2e^{\jmath t} - 2e^{-\jmath t} & 2\jmath\sin t - 2\jmath\cos t \end{bmatrix}, \\
&= e^{-t}\begin{bmatrix} \sin t + \cos t & \sin t \\ -2\sin t & -\sin t + \cos t \end{bmatrix}
\end{aligned}$$

and

$$\begin{aligned}
x(t) &= e^{At}x_0, \\
&= e^{-t}\begin{bmatrix} \sin t + \cos t & \sin t \\ -2\sin t & -\sin t + \cos t \end{bmatrix}\begin{bmatrix} v_0 \\ i_0 \end{bmatrix}, \\
&= e^{-t}\begin{bmatrix} (v_0+i_0)\sin t + v_0\cos t \\ -2v_0\sin t - i_0\sin t + i_0\cos t \end{bmatrix}
\end{aligned}$$

with $x(t) = (v_C(t), i(t))^T$ and $x(0) = x_0 = (v_0, i_0)^T$. Define ϕ such that

$$\begin{aligned}
\cos\phi &= v_0/\sqrt{2v_0^2 + 2i_0v_0 + i_0^2}, \\
\sin\phi &= -(v_0+i_0)/\sqrt{2v_0^2 + 2i_0v_0 + i_0^2}.
\end{aligned}$$

Using the identity $\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$, $v_C(t)$ can be expressed as

$$v_C(t) = \sqrt{2v_0^2 + 2i_0v_0 + i_0^2}e^{-t}\cos(t+\phi).$$

We can write

$$\begin{aligned}
i(t) &= -v_C(t) + e^{-t}((i_0+v_0)\cos t - v_0\sin t), \\
&= -v_C(t) + e^{-t}((i_0+v_0)\sin(t+\pi/2) + v_0\cos(t+\pi/2)),
\end{aligned}$$

and using the identities $\cos(\alpha+\pi/2) = -\sin\alpha$ and $\sin(\alpha+\frac{\pi}{2}) = \cos\alpha$, this can be written as

$$\begin{aligned}
i(t) &= -v_C(t) + \sqrt{2v_0^2 + 2i_0v_0 + i_0^2}e^{-t}\cos(t+\phi+\pi/2), \\
&= -v_C(t) - \sqrt{2v_0^2 + 2i_0v_0 + i_0^2}e^{-t}\sin(t+\phi), \\
&= -\sqrt{2v_0^2 + 2i_0v_0 + i_0^2}e^{-t}(\cos(t+\phi) + \sin(t+\phi)).
\end{aligned}$$

3. (a) $c_A(\lambda) = (\lambda+1)^2 + 3$. Eigenvalues are $-1 \pm \jmath\sqrt{3}$. Both are in LHP, so system is internally asymptotically stable (thus also internally stable).

   (b) $c_A(\lambda) = (\lambda-3)(\lambda+1)$. One eigenvalue in RHP, so system **not** internally stable (i.e., it is unstable).

   (c) $c_A(\lambda) = (\lambda^2+4)^2$. Eigenvalues are $\lambda_1 = 2\jmath$ with $\mathfrak{a}_{2\jmath} = 2$ and $\mathfrak{g}_{2\jmath} = 2$, and $\lambda_2 = -2\jmath$ with $\mathfrak{a}_{-2\jmath} = 2$ and $\mathfrak{g}_{-2\jmath} = 2$. So, there two repeated roots on imaginary axis, but each with geometric multiplicity equal to its algebraic multiplicity. System is internally stable but **not** asymptotically stable.

   (d) $c_A(\lambda) = \lambda^2((\lambda+1)^2+4)$. Eigenvalues are $\lambda_1 = 0$ with $\mathfrak{a}_0 = 2$ and $\mathfrak{g}_0 = 1$, $\lambda_2 = -1+2\jmath$, and $\lambda_3 = -1-2\jmath$. Since 0 is a repeated eigenvalue on imaginary axis with $\mathfrak{g}_0 < \mathfrak{a}_0$, system is unstable.

4. (a) $c_A(\lambda) = (\lambda+3)(\lambda-1)$, $\lambda_1 = -3$ and $\lambda_2 = 1$. One eigenvalue in RHP, so system is **not** internally stable (i.e., it is unstable).

   (b) No, system is not internally stable, so it cannot be internally asymptotically stable.

   (c) Controllability matrix $\mathscr{C}$ not invertible, so system not controllable.

   (d) Observability matrix $\mathscr{O}$ is invertible, so system is observable.

   (e) To determine whether the system is BIBO stable for it is initially at rest ($x_0 = 0$), we check its transfer function

$$
\begin{aligned}
H(s) &= C(sI-A)^{-1}B + D \\
&= \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} s & -3 \\ -1 & s+2 \end{bmatrix}^{-1} \begin{bmatrix} -1 \\ 1 \end{bmatrix} \\
&= \begin{bmatrix} 0 & 1 \end{bmatrix} \left(\frac{1}{s^2+2s-3} \begin{bmatrix} s+2 & 3 \\ 1 & s \end{bmatrix} \right) \begin{bmatrix} -1 \\ 1 \end{bmatrix} \\
&= \frac{\cancel{(s-1)}}{(s+3)\cancel{(s-1)}} \\
&= \frac{1}{s+3}.
\end{aligned}
$$

   Since all uncancelled poles are on the LHP, the system is BIBO stable.

   From Tutorial 2 Q5,

$$
y(t) = \frac{1}{4}[\ -e^{-3t}+e^t \quad 3e^{-3t}+e^t\ ]x_0 + \int_0^t e^{-3(t-s)}u(s)ds.
$$

Suppose u is a bounded signal: $|u(t)| \leq M$ for all $t \geq 0$. Then for $x_0 = 0$, $y(t) = \int_0^t e^{-3(t-s)} u(s) ds$ and

$$\begin{aligned} |y(t)| &\leq \int_0^t e^{-3(t-s)} M ds, \\ &= M \int_0^t e^{-3(t-s)} ds, \\ &< M e^{-3t} \int_0^t e^{3s} ds, \\ &= \frac{M}{3} e^{-3t}(e^{3t} - 1), \\ &= \frac{M}{3}(1 - e^{-3t}), \\ &\leq M/3, \end{aligned}$$

so $|y(t)| \leq N$ with $N = M/3$ for all $t \geq 0$. This gives another way to see the BIBO stability.

When $x_0 \neq (-c, c)$ for some $c \neq 0$ then $\left|\frac{1}{4}[\ -e^{-3t} + e^t \quad 3e^{-3t} + e^t\ ] x_0\right| \to \infty$ as $t \to \infty$, so also $|y(t)| \to \infty$ as $t \to \infty$, as claimed.

5. (a) $c_A(\lambda) = (\lambda + 2)(\lambda - 1)$, eigenvalues are $\lambda_1 = -2$ and $\lambda_2 = 1$. One eigenvalue in RHP, so system is internally unstable.

   (b)

$$\begin{aligned} H(s) &= C(sI - A)^{-1} B + D \\ &= \frac{s - 1}{s^2 + s - 2} \\ &= \frac{1}{s + 2}. \end{aligned}$$

   (c) From sub-problem (b) we see that the remaining uncancelled pole is on the LHP. So system is BIBO stable.

   Now, by the same calculations as in Tutorial 2 Q5, we have that (you should fully work out the details of the below)

$$e^{At} = (1/3) \begin{bmatrix} 7e^t - 4e^{-2t} & -14e^t + 14e^{-2t} \\ 2e^t - 2e^{-2t} & -4e^t + 7e^{-2t} \end{bmatrix}.$$

   and the solution for the output $y(t)$ is

$$y(t) = (1/3) e^{-2t} \begin{bmatrix} -6 & 21 \end{bmatrix} x_0 + \int_0^t e^{-2(t-s)} u(s) ds.$$

Note that none of the terms on the right hand side contain the term e^t, eventhough this is an eigenvalue of the system that causes instability. This is due to the system being unobservable. Since the system is BIBO stable, $\int_0^t e^{-2(t-s)}u(s)ds$ is bounded whenever u is a bounded input, and the term $(1/3)e^{-2t}\begin{bmatrix} -6 & 21 \end{bmatrix} x_0$ is obviously bounded for any time $t \geq 0$. So, $y(t)$ is a bounded signal whenever the input signal $u(t)$ is bounded, despite the system being internally unstable. Unobservability of the system can hide internal instabilities.